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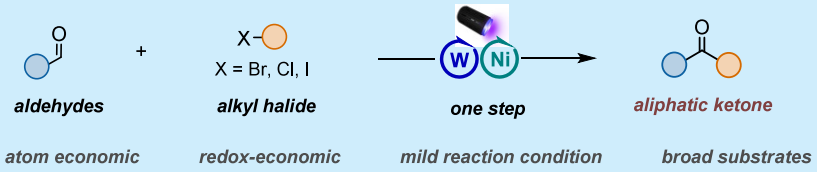
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A General Direct Aldehyde C−H Alkylation via TBADT-Nickel Synergistic Catalysis

Zhimin Hu,† Ji-Wei Sang,† Shize Xie,† Yujia Song,† Qiuhao Li, Bing Wang, Zheng-Yang Xu, Yu Zhang, Runhui Liu,* Wei-Dong Zhang,* and Jinxin Wang*

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ABSTRACT: Herein, we present a nickel/tetrabutylammonium decatungstate (TBADT)-catalyzed protocol for general C−H alkylation of aldehydes, enabling the efficient synthesis of aliphatic ketones through ligand-controlled cross-coupling. This mild and cost-effective methodology demonstrates broad substrate compatibility with various commercially available aldehydes and both activated and unactivated alkyl bromides, delivering target products in high yields. Notably, the practical utility of this catalytic system has been highlighted through a concise two-step synthesis of the commercially valuable musk odorant Aurelione from readily available starting materials.

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Developments in new synthetic methods continue to have an impact on the discovery and development of the next generation of medicines.[1] Aliphatic ketones are kinds of skeletons that frequently appear in natural products, pharmaceuticals, and flavors, such as prasugrel, methadone, trichostatin A, myriocin, muscone, and Aurelione (Figure 1).

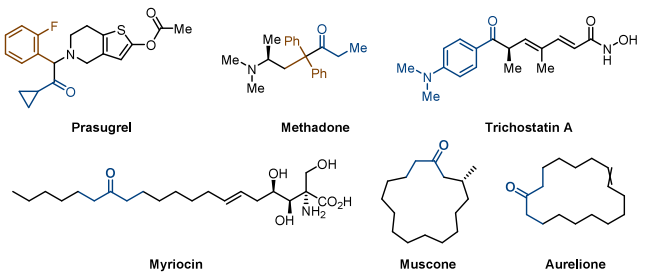


Figure 1. Selected examples of aliphatic ketones from natural products or drugs with biological activities.

They often serve as the key intermediates for the synthesis of various pharmaceutical skeletons, for instance, chiral alcohol, chiral amine, indole, imidazole, quinoline, and other heterocyclic skeletons.[2] Although traditional nucleophilic addition of organometallic reagents to carbonyl moieties, including aldehyde, amides, anhydrides, and acid chlorides, or transition-metal-catalyzed cross-coupling of acyl electrophiles (e.g., acyl halides, anhydrides, esters, and amides) with organometallics or less reactive carbon nucleophiles are among the most widely used reactions in pharmaceutical and synthetic chemistry,[3] the employment of (super)stoichiometric amounts of organometallics reagents limits the synthetic practicability and functional group compatibility and also does not meet sustainability standards. For these reasons, developing novel and efficient methods toward the synthesis of ketones has been a concern for the organic synthesis community (Scheme 1).

The reductive cross-coupling of two shelf-stable and readily accessible electrophiles catalyzed by nickel offers a more efficient and selectively controllable pathway organic chemical synthesis method under mild conditions to overcome the high energy barrier steps encountered in traditional transformation by alkyl radical addition processes[4,5] (Scheme 1a). Recently, metallaphotoredox has enabled the development of a wide variety of cross-coupling reactions.[6] For forging C(acyl)−C(sp^3) bonds,[7] the alkyl radical can be accessed from alkyltrifluoroborates,[7c−e] alkyl bromides,[7m] activated esters,[7a,q] 1,4-dihydropyridines,[7g,i,j,n] alcohols,[7s] and alkanes[7b,f,h,l,o,p] by photoredox in the presence of a photocatalyst and organic reductants or an exogenous source of halide anions instead of the aforementioned metal reductants (Scheme 1b). An alternative yet equally attractive approach to ketone formation has emerged that involves direct C−H functionalization of aldehydes enabled by metallaphotoredox. This strategy takes

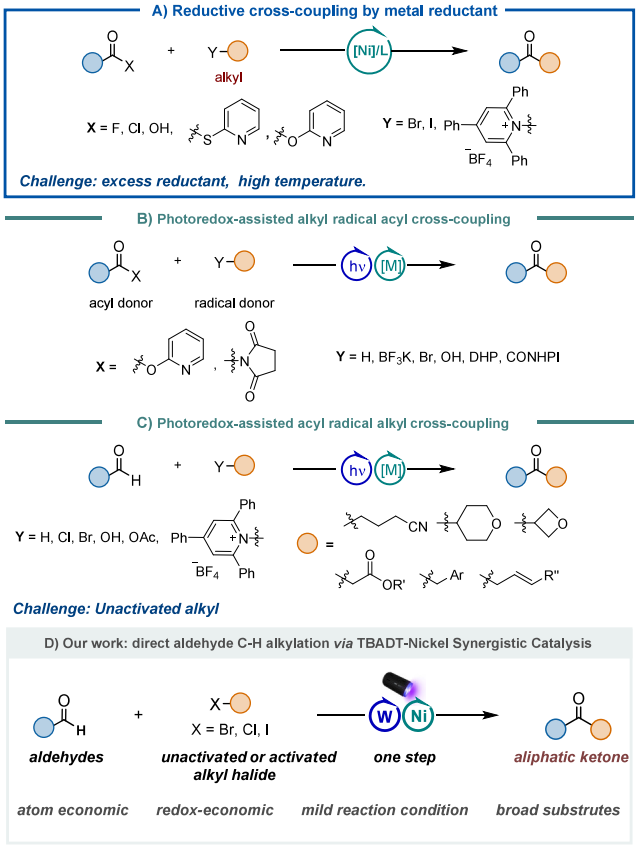
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Scheme 1. Ketone Synthesis Based on Transition-Metal-Catalyzed Cross-Coupling

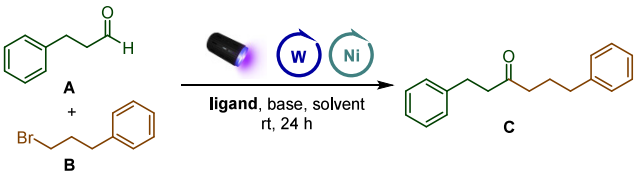


advantage of high regioselectivity toward formyl C−H bonds even in the presence of various C−H bonds with similar bond dissociation energies.[8] In this system, preactivation of the acyl group is not required, and the production of the radical does not require additional oxidants or reducing agents. The seminal work was successfully accomplished by MacMillan[9] via synergistic combination of iridium photoredox, nickel, and hydrogen atom transfer catalysis in a triple-catalytic activation mechanism that allows aryl, vinyl, or alkyl aldehydes to undergo direct C−H arylation and alkylation. For the benzylation, Murakami and Ishida[10] developed a dehydrogenative coupling of aldehydes and alkylbenzenes by Ir/Ni dual catalysis. As an alternative to the use of an expensive noble-metal photocatalyst, arylations and alkenylations of aldehydes were achieved by merging tetrabutylammonium decatungstate (TBADT) with a nickel or palladium catalyst. Different activated alkylations also were disclosed. Wang et al.[11] successfully employed different carbon nickel intermediates to couple with acyl radicals from aldehydes to construct the $C(sp^2)-C(sp^3)$ bond. Rasappan et al.[12] demonstrated a milder and efficient method for the cross-coupling of alkyl and aryl aldehydes with benzylic and allylic pyridinium salts via C−N bond cleavage for the first time (Scheme 1c). However, the substrate scope of direct C−H alkylation of aldehydes is still limited (only three examples), and a general and practical approach to achieve aliphatic ketones is still in high demand.

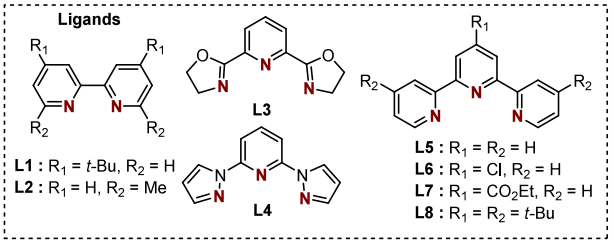
Our group focuses on developing practical reactions to synthesize and modify bioactive natural products to discover new drugs. Following up on our previous works,[13] we aimed to develop direct aldehyde C−H functionalization reactions to obtain various ketones efficiently and conveniently. Herein we report a general aliphatic ketone synthesis method using nickel/TBADT synergistic-catalyzed coupling of aldehydes with alkyl halides via ligand control.

We initiated our investigation by coupling phenylpropanal **A** with unactivated alkyl bromides **B** to access ketone products **C** under the merged TBADT and nickel catalyst, and the results are summarized in Table 1 and Table S1 in the Supporting Information). In contrast to the known reports on cross-electrophile couplings, the frequently used 2,2′-bipyridine ligands such as electron-rich dtbbpy (**L1**) and di(Me)bpy (**L2**) were did not accomplish the desired transformation, where primarily the alkyl bromide was recovered. A bipyridine ligand with steric encumbrance at the 6,6′ positions showed the same inefficiency. We noted that coordination of a heteroatom in the alkyl bromide with nickel might promote the generation of an acyl−alkylnickel(III) bromide complex via an oxidative addition or "radical rebound" mechanism.[10,14,15] So, we supposed that the proper ligand could provide the same effect. Inspired by the clear and comprehensive computational analysis of the mechanisms of the photoredox-nickel-HAT-catalyzed cross-coupling reactions[15] and the ability of the terpyridine ligand to stabilize alkyl complexes of nickel in

Table 1. Optimization of Aldehyde C−H Alkylation



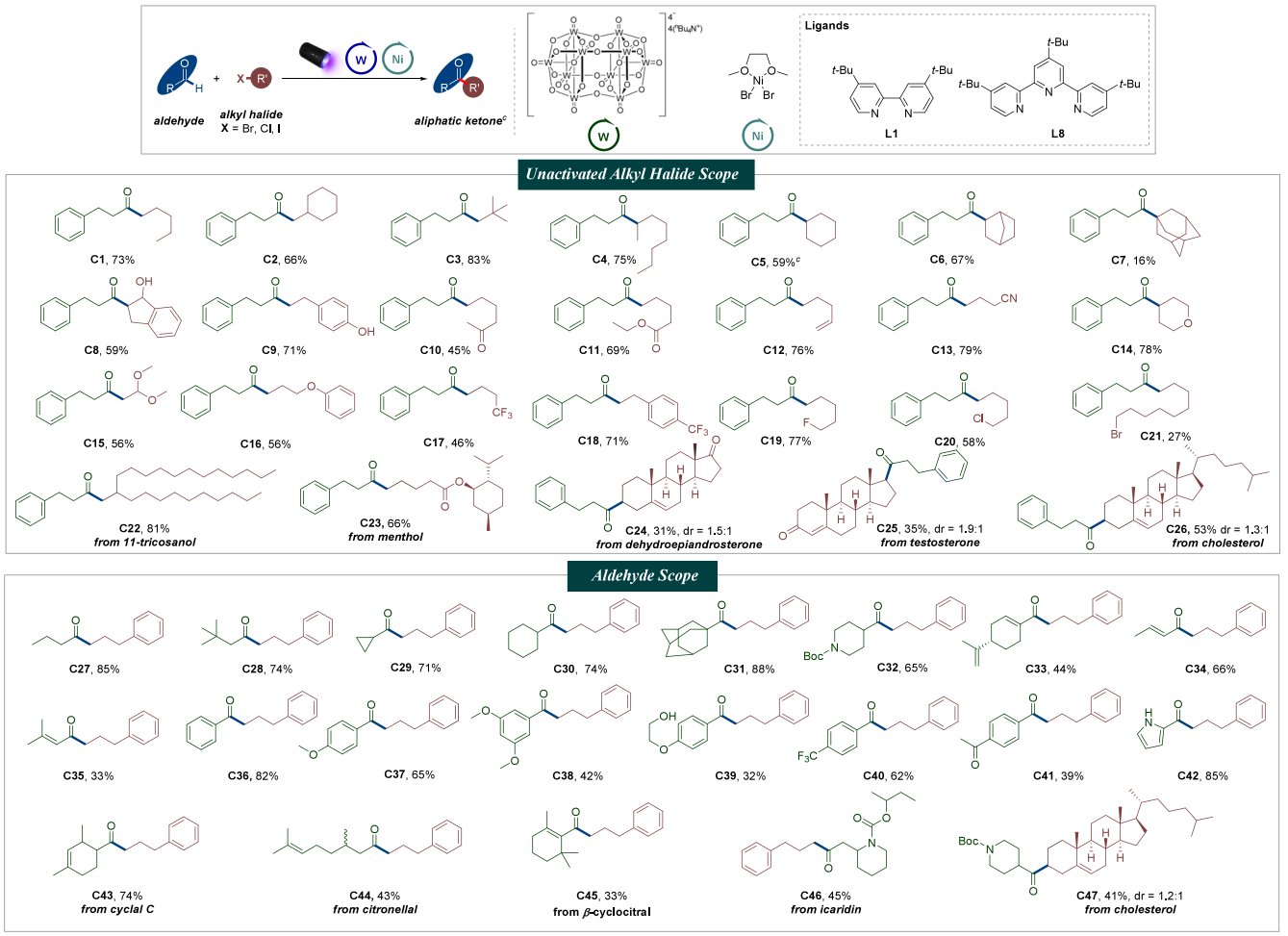
Entry[a]	Reaction conditions	Yield[b] (%)
1	**L1**	trace
2	**L2**	trace
3	**L3**	15
4	**L4**	9
5	**L5**	24
6	**L6**	31
7	**L7**	29
8	**L8**	45
9	**L8**, concentration (0.1 M)	62
10	**L8**, concentration (0.2 M)	76
11	**L8**, concentration (0.3 M)	72
12[c]	**no** TBADT	0
13[c]	no $NiBr_2$•glyme	0
14[c]	no ligand	0
15[c]	no light	0



[a]Reaction conditions: TBADT (2 mol%), $NiBr_2$·glyme (10 mol%), ligand (10 mol%), alkyl halide (1.0 equiv), aldehyde (3.0 equiv), Na_2CO_3 (3.0 equiv), acetone (0.05 M), 40 W LED lamp (390 nm) at rt with cooling fan under argon atmosphere for 24 h. [b]Isolated yield. [c]**L8**, acetone (0.2 M).

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Scheme 2. Substrate Scope for the Coupling of Alkyl Halides and Aldehydes



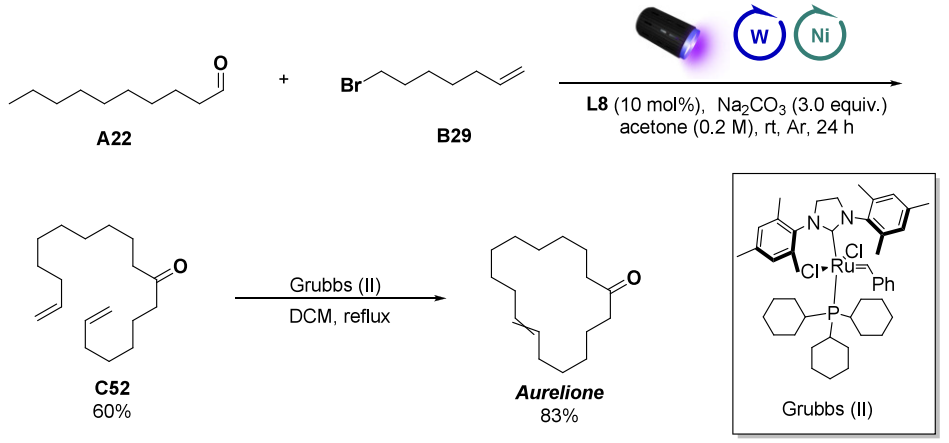
[a]TBADT (2 mol%), $NiBr_2$·glyme (10 mol%), **L8** (10 mol%), alkyl halide (1.0 equiv), aldehyde (3.0 equiv), Na_2CO_3 (3.0 equiv), acetone (0.2 M), 40 W LED lamp (390 nm) at rt with cooling fan under argon atmosphere for 24 h. [b]Yield of the isolated product after column chromatography. [c]Halide substrate is iodide.

alkyl−alkyl cross-coupling reactions,[16] a variety of tridentate ligands were examined, and 24%−45% yields were obtained (ligands **L3**−**L8**, Table 1, entries 3−8, and Table S1 in the Supporting Information). Notably, the ligand **L8** showed the best results. After systematic screening of different bases, solvents (Table S1 in the Supporting Information), and reaction concentrations (Table 1, entries 9−11), we found optimal conditions for coupling an aldehyde with an unactivated alkyl halide (Table 1): TBADT (2 mol%), $NiBr_2$·glyme (10 mol%), **L8** (10 mol%), Na_2CO_3 (3.0 equiv), and acetone (0.2 M) with exposure to 390 nm LEDs at ambient temperature. These conditions provided the aliphatic ketone product **C** in 76% isolated yield (entry 11, Table 1). Control experiments (entries 12−15) revealed that the photocatalyst, nickel catalyst, ligand, and irradiation were all essential for the success of this new aldehyde C−H alkylation.

With the optimized conditions, we next sought to determine the generality of the alkyl halide component in this new aliphatic ketone-forming reaction. The scope of these conditions is broad, as demonstrated by the 47 examples in Scheme 2. For the unactivated alkyl bromide, linear alkyl bromides, branched chain alkyl bromides, and cyclic and bridge-cyclic alkyl bromides were able to functionalize phenylpropyl aldehyde and afford various ketones in moderate to good yields (**C1**−**C6**, 59%−83% yield). Caged tertiary alkyl bromide was also used (**C7**, 16%). Under mild reaction conditions and with the high selectivity of TBADT for aldehyde, a variety of functional groups were well tolerated (**C8**−**C18**, 45%−79%). Importantly, substrates bearing hydroxyl, phenolic hydroxyl, and ketone carbonyl groups, which are incompatible in Grignard reactions and in cross-coupling reactions with metal reductants or active acyl donors, can be readily employed in this Ni/TBADT metallaphotoredox aldehyde direct alkylation. Groups such as esters, nitriles, alkenyls, cyclic ethers, phenol ethers, acetals, and both aliphatic and aryl trifluoromethyl delivered the products in acceptable to good yields. Through chemoselective cross-coupling of the bromide motif to aldehydes, the construction of halogen-substituted aliphatic ketones could effectively reduce the number of reaction steps (**C19**−**C20**, 77% and 58% yield, respectively), and 1,9-dibromnonan gave the homocoupling product in an acceptable yield (**C21**, 27%). Notably, these versatile products provide additional handles for further elaboration and diversification. To examine the potential for late-stage application of this coupling reaction, alkyl bromides derived from natural product or drugs, 11-tricosaniol (**C22**), (−)-menthol (**C23**), dehydroepiandrosterone (**C24**), testos-

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Scheme 3. Two-Step Synthesis of Aurelione



terone (**C25**), and cholesterol (**C26**), successfully underwent coupling with phenylpropanal in acceptable to excellent yields (31%−81%). Activated alkyl bromides (benzyl, allyl, propargyl) served as effective substrates in cross-coupling with bidentate ligand **L1** in dilute solution (0.05 M), affording products in up to 98% yield (Tables S2 and S3 in the Supporting Information). Electron-rich benzyl bromides containing alkyl, aryl, and methoxy groups performed well (**C54**−**C59**, 81%−95% yield). Moreover, a variety of electron-deficient benzyl bromides or chlorides containing groups such as trifluoromethoxy, nitro, sulfones, esters, nitriles, and trifluoromethyl, fluoro, and chloro groups were found to be effective electrophiles (**C60**−**C68**, 52%−93% yield). In addition to benzene rings, naphthyl, quinolone, and furan groups were suitable substrates (**C69** and **C71**, 91% and 78% yield, respectively). Notably, the efficiency of the reaction was not impeded by ortho substituents on the aromatic ring (**C63**, **C66**, and **C68**) or by using secondary benzyl halides (**C70**, 96% yield). Perhaps most importantly, this transformation is not limited to benzyl halides; for example, allyl halides and propargyl halides can be readily employed in this alkylation (**C72**−**C74**, 59%−88% yield).

We next turned our attention to the scope of the aldehyde component. As shown in Scheme 2, an assortment of aliphatic acetones could be obtained in good to excellent yields with a broad range of aldehydes. Aliphatic aldehydes including linear, branched-chain, and cyclic primary, secondary, and even tertiary aldehydes were effective coupling partners (**C27**−**C31** and **C75**−**C77**, **C82**−**C84** in the Supporting Information, 47%−94% yield). Moreover, aliphatic aldehydes containing *N*-Boc-piperidine, phthalimide, isopentyl, tertiary alcohols, and furan were also coupled with high levels of efficiency (**C32** and **C78**−**C81** in the Supporting Information, 61%−98% yield). Strikingly, α,β-unsaturated aldehydes were successfully subjected to cross-coupling for the first time, affording α,β-unsaturated ketones (**C33**−**C35** and **C93**−**C95** in the Supporting Information, 59%−87% yield) in moderate to excellent yields. This achievement is particularly significant given the inherent challenges in suppressing competing Giese addition pathways, a critical concern arising from the inherent electrophilicity of both the enal substrates and the resultant enone products. Lastly, aromatic aldehydes were found to couple with alkyl halides proficiently, despite the diminished hydridic nature of these C−H bonds. With the exception of unsubstituted benzaldehydes (**C36** and **C85** in the Supporting Information, 82% and 94% yield), electron-donating groups such as alkoxy and methyl were eligible substituents on the aromatic ring (**C37**−**C39** and **C88**−**C89** in the Supporting Information, 32%−93% yields). Electron-withdrawing groups, including common substituents such as trifluoromethyl, acetyl, and nitro groups, could be coupled with this method (**C40**−**C41** and **C86**−**C87** in the Supporting Information, 39%−62% yields). With respect to heteroaromatic coupling partners, we found that a range of substituted heteroaryl aldehydes are also effective substrates (**C42** and **C90**−**C92** in the Supporting Information, 85%, 86%, 63%, and 46% yield, respectively). Perhaps more important was the finding that aldehydes from natural products or drugs can be successfully employed to generate ketones (**C43**−**C47**, 33%−74% yield). We believe that these results demonstrate the first broadly general approach to access aliphatic ketones from aldehydes and alkyl halides.

To highlight a potential application to valuable molecules, we accomplished a two-step synthesis of Aurelione, a well-established musk odorant.[17] As shown in Scheme 3, 10-undecenal **A22** and 7-bromo-1-heptene **B29** were successfully combined using our aldehyde coupling protocol to forge ketone **C52** in a good yield (60%). Exposure of the aforementioned octadeca-1,17-dien-8-one to the Grubbs catalyst (II) in dichloromethane under reflux resulted in the formation of Aurelione in 83% yield.

In summary, we have developed the first photoinduced nickel-catalyzed cross-coupling protocol enabling direct conjugation of aldehydes with diverse alkyl halides. This sustainable methodology features operationally simple and safe procedures conducted under environmentally benign conditions, demonstrating excellent functional group compatibility while eliminating the reliance on precious metal catalysts. The synthetic utility of this strategy has been demonstrated through an efficient two-step synthesis of the commercially valuable musk fragrance Aurelione from readily available starting materials. This catalytic paradigm not only establishes a practical platform for sustainable aliphatic ketone synthesis but also significantly expands accessible chemical space for medicinal chemistry exploration while addressing critical challenges in an environmentally conscious synthetic design.

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■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c00949.

Experimental procedures, condition optimization, the substrate scope of the activated alkyl bromides and aldehydes, mechanism experiments, crystal data, and NMR spectra (PDF)

Accession Codes

Deposition Number 2160760 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

■ AUTHOR INFORMATION

Corresponding Authors

Runhui Liu − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*; Email: lyliurh@126.com

Wei-Dong Zhang − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China; School of Pharmaceutical Sciences, Wenzhou Medical University, Wenzhou 325035 Zhejiang, China; Asymchem Laboratories (Tianjin) Co., Ltd., Tianjin 300462, China; Institute of Interdisciplinary Integrative Medicine Research, Shanghai University of Traditional Chinese Medicine, Shanghai 201203, China; State Key Laboratory for Quality Ensurance and Sustainable Use of Dao-di Herbs, Institute of Medicinal Plant Development, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing 100193, China*; Email: wdzhangy@hotmail.com

Jinxin Wang − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*; orcid.org/0000-0001-9561-7633; Email: jxwang2013@126.com

Authors

Zhimin Hu − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*

Ji-Wei Sang − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*

Shize Xie − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*

Yujia Song − *School of Pharmaceutical Sciences, Wenzhou Medical University, Wenzhou 325035 Zhejiang, China*

Qiuhao Li − *Asymchem Laboratories (Tianjin) Co., Ltd., Tianjin 300462, China*

Bing Wang − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*

Zheng-Yang Xu − *Department of Phytochemistry, School of Pharmacy, Second Military Medical University, Shanghai 200433, China*; orcid.org/0000-0003-4605-8286

Yu Zhang − *Asymchem Laboratories (Tianjin) Co., Ltd., Tianjin 300462, China*; orcid.org/0000-0002-7230-3575

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.5c00949

Author Contributions

†Z. H., J.-W. S., S. X., and Y. S. contributed equally.

Notes

The authors declare no competing financial interest.

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